

AWS Platform Engineer

3-Year Experience Interview Handbook

End-to-End Questions • Model Answers • Architecture • Scenarios •
Troubleshooting • Mock Interview

Candidate positioning: 3+ years DevOps / cloud-native experience with AWS, EKS, Jenkins, ArgoCD, Helm, Terraform, Ansible, Prometheus/Grafana, security tooling and production-style microservices platforms.

Designed for a senior interviewer / AWS Architect-style second round.

Goal: answer like a practical platform engineer — explain the design, state the trade-off, show the troubleshooting path, and be honest about the depth of your hands-on exposure.

How to Use This in the Next 24 Hours

0–3 hours: VPC, networking, IAM, ALB/NLB, Route 53, NAT, VPC endpoints.

3–6 hours: EKS architecture, node groups, networking, IAM integration, ingress/load balancing, storage.

6–9 hours: EC2, Auto Scaling, RDS, S3, EBS/EFS, CloudWatch, CloudTrail, KMS, Secrets Manager.

9–12 hours: Terraform + AWS, CI/CD + AWS, security, cost, HA/DR, multi-AZ design.

12–16 hours: Work through all scenario questions and troubleshooting flows aloud.

16–20 hours: Mock interview: answer without looking, then compare to model answers.

20–22 hours: Rapid-fire revision tables and commands.

22–24 hours: Sleep/rest and final 60-minute revision. Do not learn new services at the end.

Interview answer formula

Use: What → Why → How → Trade-off → Example. **For troubleshooting use: symptom → scope → recent change → dependency chain → logs/metrics → isolate → fix → prevent recurrence.**

1. Your AWS Platform Engineer Story

Keep your introduction aligned to what you can defend technically. Do not claim ownership of AWS services you have only studied.

Good morning. I'm Srinithin. I have more than three and a half years of experience in DevOps, primarily across cloud infrastructure, CI/CD automation, containerization and application deployments. My recent work has been around cloud-native microservices running on AWS EKS, with Jenkins for CI, ArgoCD and Helm for GitOps-based deployment, Terraform for infrastructure automation, and Prometheus, Grafana and CloudWatch for monitoring. From the AWS side, my strongest areas are VPC networking, IAM, EC2, load balancing, EKS, ECR, storage and operational troubleshooting. I have worked in production-style environments where availability, security, deployment automation and observability were important, and I focus on solving issues systematically rather than only operating individual tools.

Q: Why are you moving toward an AWS Platform Engineer role?

A: My experience has increasingly moved toward infrastructure automation, Kubernetes platforms, CI/CD, cloud networking, security and operational reliability. A platform engineering role is a natural progression because it combines those areas into building reusable, secure and self-service infrastructure for application teams.

Q: What is the difference between DevOps Engineer and Platform Engineer?

A: DevOps focuses heavily on delivery and operational collaboration. Platform engineering builds the internal platform that makes delivery repeatable: infrastructure, Kubernetes, CI/CD templates, identity, observability, security guardrails and developer self-service. In practice there is overlap, but the platform engineer thinks more in terms of reusable capabilities and standards.

Q: What would you own as an AWS Platform Engineer?

A: I would expect ownership of AWS infrastructure patterns, VPC/networking, EKS platform, IAM integration, load balancing, observability, security controls, Terraform modules, CI/CD integration, reliability and cost optimization, while application teams own their application code and business logic.

2. AWS VPC & Networking — High Priority

Q: Explain a production VPC design.

A: I would use multiple AZs, public and private subnets, route tables per subnet tier, Internet Gateway for public connectivity, NAT Gateway for private outbound traffic, and VPC endpoints where appropriate. Internet-facing ALBs sit in public subnets; application and EKS nodes normally stay private. Databases stay private and are reachable only from approved application security groups.

Q: What makes a subnet public?

A: A subnet is considered public when its route table has a default route to an Internet Gateway. Public IPv4 addressing is a separate requirement for direct internet reachability of an individual resource.

Q: Explain route table evaluation.

A: The most specific matching route is selected. A default route such as 0.0.0.0/0 is used when no more specific route exists. Route tables are associated with subnets.

Q: IGW vs NAT Gateway?

A: An Internet Gateway provides VPC internet connectivity and supports public resources. NAT Gateway allows resources in private subnets to initiate outbound internet connections without accepting unsolicited inbound internet connections.

Q: Why NAT Gateway per AZ?

A: For resilience and to avoid dependence on another AZ. A NAT Gateway is AZ-scoped. One per AZ avoids cross-AZ dependency and can also reduce cross-AZ data transfer.

Q: Security Group vs NACL?

A: Security Groups are stateful, resource/ENI-level controls and support allow rules. NACLs are stateless subnet-level controls with both allow and deny rules. I use SGs for primary workload access control and NACLs as an additional subnet boundary.

Q: What is VPC peering?

A: Private connectivity between two VPCs using private IPs. It is non-transitive, so it becomes difficult to manage at scale. For many VPCs, Transit Gateway is generally a better centralized pattern.

Q: What is Transit Gateway?

A: A regional network transit hub used to connect multiple VPCs and on-premises networks. It simplifies routing compared with a large mesh of VPC peerings.

Q: What is a VPC endpoint?

A: Private connectivity from a VPC to supported AWS services without routing traffic through the public internet. Gateway endpoints are commonly used for S3 and DynamoDB; interface endpoints use private ENIs and PrivateLink.

Q: Why use an S3 VPC endpoint instead of NAT?

A: It keeps S3 access on the AWS private network and can reduce NAT Gateway data-processing cost. It also removes an unnecessary internet egress dependency.

Q: What is CIDR?

A: Classless Inter-Domain Routing defines an IP range such as 10.0.0.0/16. In VPC design I allocate non-overlapping CIDRs so VPCs, subnets, peering and Transit Gateway routing remain manageable.

Q: What is DNS resolution in a VPC?

A: VPC DNS support and DNS hostnames allow resources to resolve AWS service and internal names. Route 53 private hosted zones can provide internal DNS names.

Scenario: EKS pods cannot reach the internet

Scenario: Pods are healthy but package downloads and external API calls fail.

Interview approach: Trace the dependency chain: pod → node ENI → private subnet route table → NAT Gateway → public subnet route → Internet Gateway. Check the private route table, NAT state, NAT subnet route, NACLs, security groups and DNS. If the destination is an AWS service, check whether a VPC endpoint is more appropriate.

Useful checks:

```
kubectll get pods -A
kubectll exec -it <pod> -- sh
# inside pod:
nslookup example.com
curl -I https://example.com
# AWS-side:
aws ec2 describe-route-tables
aws ec2 describe-nat-gateways
```

Scenario: Application in private subnet cannot connect to RDS

Scenario: The application resolves the RDS endpoint but connection times out.

Interview approach: First verify DNS and port reachability. Then inspect the RDS security group inbound rule, application security group, subnet routes, NACLs and RDS state. Prefer SG-to-SG rules rather than broad CIDR rules. Confirm the database port and whether the RDS endpoint is the correct writer/reader endpoint.

Useful checks:

```
nslookup <rds-endpoint>
nc -zv <rds-endpoint> 5432
kubectll exec -it <pod> -- nc -zv <rds-endpoint> 5432
```

3. EC2, ALB, NLB & Auto Scaling

Q: What is an AMI?

A: A machine image used to launch EC2 instances. It contains the operating system and can include software/configuration. Immutable AMIs make fleet provisioning more repeatable.

Q: How do you troubleshoot an unreachable EC2 application?

A: Check instance/system status, process status, listening socket, local curl, security group, NACL, route table, public/private connectivity, load balancer target health and application logs.

Q: ALB vs NLB?

A: ALB is Layer 7 and supports HTTP/HTTPS features such as host/path routing. NLB is Layer 4 and is designed for TCP/UDP/TLS workloads, high throughput and low latency.

Q: What is a target group?

A: A logical set of targets behind a load balancer with health-check configuration and routing behavior.

Q: Why is an ALB target unhealthy?

A: Common causes are wrong health-check path/port/protocol, security-group restrictions, application not listening, readiness/application failure, incorrect target registration or response outside the expected health-check codes.

Q: What is Auto Scaling Group?

A: A controller for an EC2 fleet that maintains desired capacity and can scale between minimum and maximum based on policies or schedules.

Q: How would you achieve zero-downtime EC2 deployment?

A: Use an immutable or rolling deployment pattern with an ALB and ASG, launch healthy new instances, register them, wait for health checks, then drain and terminate old instances.

Q: What is instance profile?

A: An IAM role attached to an EC2 instance through an instance profile, allowing applications on the instance to obtain temporary AWS credentials.

4. EKS — Highest Priority for Your Profile

Q: Explain EKS architecture.

A: AWS manages the Kubernetes control plane, including the API server and etcd. The customer manages worker capacity, cluster add-ons, networking configuration, IAM integration, workloads, policies and operations. EKS can use managed node groups, self-managed nodes and serverless/Fargate options depending on workload.

Q: Where is kubeconfig stored after aws eks update-kubeconfig?

A: Normally in ~/.kube/config on the machine where the command is executed, unless KUBECONFIG or an explicit --kubeconfig path is used. It is not automatically stored on worker nodes.

Q: How does kubectl authenticate to EKS?

A: The kubeconfig invokes AWS CLI authentication to obtain a token/credential flow, which EKS maps to Kubernetes authorization through the cluster's access configuration. Authorization is then controlled by Kubernetes RBAC and/or EKS access policies.

Q: How does an EKS pod access S3 securely?

A: Use EKS Pod Identity or IRSA to associate a narrowly scoped IAM role with the workload. Avoid giving every worker node broad S3 permissions.

Q: What is ECR?

A: AWS managed container image registry. EKS nodes need authorization to pull images, and the cluster must have network connectivity to ECR endpoints.

Q: Why can a pod be Running but not receive traffic?

A: Running is a lifecycle state. A pod may fail readiness checks, have no matching Service endpoints, or have an application/port issue. Traffic should normally be gated by readiness.

Q: What is a Service?

A: A stable virtual endpoint that selects pods. ClusterIP is internal, NodePort exposes a port on nodes, and LoadBalancer integrates with cloud load-balancing behavior.

Q: Ingress vs Service?

A: Service provides stable network access to pods. Ingress defines HTTP/HTTPS routing rules and is implemented by an ingress controller such as an AWS load-balancer controller.

Q: How does an AWS ALB reach EKS pods?

A: The AWS Load Balancer Controller watches Kubernetes resources and provisions/configures AWS load-balancing resources. Depending on target mode, traffic can target node ports or pod IPs.

Q: EBS CSI vs EFS CSI?

A: EBS CSI provides block storage, commonly for RWO-style workloads. EFS CSI provides shared filesystem access and supports RWX-style use cases across nodes/AZs.

Q: What happens if an EKS node fails?

A: Pods on the failed node become unavailable and the scheduler/controller mechanisms replace rescheduled replicas on healthy nodes if capacity and scheduling constraints allow it. Stateful workloads depend on their storage and topology configuration.

Q: How do you upgrade EKS safely?

A: Check version compatibility and add-ons, update control plane, then node groups, respecting supported version skew and disruption budgets. Test in non-production, use rolling node replacement and monitor workloads throughout.

Q: How do you debug CrashLoopBackOff?

A: Inspect pod description/events, previous container logs, current logs, command/args, environment variables, mounted secrets/config, probes, resource limits and application startup behavior.

Q: How do you debug ImagePullBackOff?

A: Check image name/tag, ECR repository, IAM pull permissions, node/pod identity, network/DNS access to ECR, registry authentication and whether the image exists in the expected region.

Scenario: EKS deployment causes 503 errors

Scenario: Pods appear Running, but ALB returns 503 after a release.

Interview approach: Check ALB target health, Service selectors/endpoints, pod readiness, container port versus Service targetPort, ingress annotations, security groups and application logs. Compare with the last known-good deployment and roll back if the release is clearly the cause.

Useful checks:

```
kubectl get deploy,pods,svc,endpoints -n <ns>
kubectl describe pod <pod> -n <ns>
kubectl get ingress -n <ns>
kubectl get events -n <ns> --sort-by=.lastTimestamp
```

Scenario: EKS nodes are NotReady

Scenario: Multiple workloads suddenly fail.

Interview approach: Check node conditions, kubelet, container runtime, disk/memory/PID pressure, network connectivity to the control plane, IAM and CNI health. Determine whether the issue is one node or a common dependency such as networking or capacity.

Useful checks:

```
kubectl get nodes
kubectl describe node <node>
kubectl get pods -n kube-system
journalctl -u kubelet --no-pager
systemctl status containerd
```

5. IAM, Security & Secrets

Q: User vs role?

A: A user is a long-lived identity; a role is assumed and provides temporary credentials. Workloads should generally use roles rather than embedded access keys.

Q: What is least privilege?

A: Grant only the actions and resources required for a workload, with conditions where possible. Review and reduce permissions over time.

Q: Identity policy vs resource policy?

A: Identity policies attach to identities such as users/roles. Resource policies attach to resources such as S3 buckets or KMS keys and can enable cross-account access.

Q: How do you secure production AWS credentials?

A: Avoid static credentials in code and CI logs. Use IAM roles, workload identity, Secrets Manager/Parameter Store where appropriate, KMS encryption, short-lived credentials and auditing.

Q: Secrets Manager vs Parameter Store?

A: Both store configuration/secrets. Secrets Manager is designed for secrets management and supports features such as rotation; Parameter Store is useful for configuration parameters and can also store encrypted SecureString values.

Q: What is KMS?

A: AWS managed key-management service used to create and control encryption keys and integrate encryption into services such as S3, EBS, RDS and Secrets Manager.

Q: What is CloudTrail?

A: An audit service that records AWS API activity. It is used for security investigation, governance and change attribution.

Q: What is GuardDuty?

A: A managed threat-detection service that analyzes relevant AWS data sources for suspicious activity.

Q: How do you secure an EKS cluster?

A: Private networking where appropriate, restrictive security groups, IAM least privilege, workload identity, Kubernetes RBAC, network policies, admission policies, image scanning/signing, encrypted secrets, audit logging, patching and continuous monitoring.

Scenario: A developer accidentally exposed an AWS access key

Scenario: An access key was committed to Git.

Interview approach: Treat it as compromised immediately: deactivate/rotate the key, investigate CloudTrail for use, identify blast radius, remove it from source history where appropriate, replace it with role-based authentication, and add secret scanning/pre-commit/CI controls. Do not simply delete the Git file and assume the credential is safe.

Useful checks:

```
aws iam list-access-keys --user-name <user>
aws cloudtrail lookup-events --lookup-attributes AttributeKey=Username,AttributeValue=<user>
# Then rotate/revoke according to the incident procedure.
```

6. S3, RDS, EBS, EFS & Data Services

Q: S3 object storage vs EBS block storage?

A: S3 stores objects through an API and is highly durable object storage. EBS provides block devices attached to compute and is suited to filesystems and databases.

Q: EBS vs EFS?

A: EBS is block storage and typically tied to an AZ; EFS is a managed regional network filesystem designed for shared access across multiple compute instances/AZs.

Q: RDS Multi-AZ vs Read Replica?

A: Multi-AZ is primarily high availability/failover. Read replicas are primarily read scaling and can also support some DR patterns depending on architecture.

Q: How do you protect an RDS database?

A: Private subnets, restrictive security groups, encryption with KMS, automated backups, appropriate retention, deletion protection where required, monitoring, patching and least-privilege database access.

Q: What is S3 lifecycle policy?

A: Rules that transition or expire objects based on age/tags/prefix and storage requirements, reducing cost and automating retention.

Q: What is S3 versioning?

A: Maintains multiple object versions, enabling recovery from accidental overwrite/deletion.

Q: What is S3 encryption?

A: Objects can be encrypted using S3-managed keys or AWS KMS-backed encryption. For sensitive workloads, KMS provides stronger key-management control and auditing.

7. Monitoring, Logging & Reliability

Q: CloudWatch vs Prometheus?

A: CloudWatch is AWS-native monitoring/logging for AWS resources and services. Prometheus is a metrics system commonly used for Kubernetes/application metrics. They complement each other.

Q: What would you monitor for EKS?

A: Node CPU/memory/disk/PID pressure, pod restarts, pending pods, API-server/control-plane signals available to the environment, CNI health, application latency/errors, ALB target health, cluster capacity and deployment health.

Q: Logs vs metrics vs traces?

A: Metrics quantify system behavior over time, logs provide event/detail context, and traces follow a request across distributed services. In microservices, using all three improves incident diagnosis.

Q: What is an SLI/SLO?

A: An SLI is a measured indicator such as availability or latency. An SLO is the target level for that indicator. Error budgets turn reliability targets into an operational decision framework.

Q: How do you design alerts?

A: Alert on user-impacting symptoms and actionable causes, use thresholds/percentiles appropriate to the metric, avoid noisy alerts, include ownership/runbook context and distinguish warning from critical.

Scenario: CPU is 90% on EKS nodes

Scenario: Several nodes show sustained high CPU and pods are becoming pending.

Interview approach: First establish whether the load is expected or anomalous. Check pod requests/limits, top CPU consumers, HPA behavior, node capacity and scheduling constraints. Scale workloads if appropriate; add node capacity if the cluster is genuinely under-provisioned. Then identify the application cause so scaling does not become the permanent fix.

Useful checks:

```
kubectl top nodes
kubectl top pods -A --sort-by=cpu
kubectl get hpa -A
kubectl get nodes
kubectl describe node <node>
```

8. Terraform + AWS Platform Engineering

Q: Why Terraform?

A: It provides declarative, version-controlled infrastructure provisioning with reusable modules and predictable change plans.

Q: What is Terraform state?

A: A mapping between configuration and managed infrastructure. It allows Terraform to determine what exists and what changes are required.

Q: Why remote state?

A: For collaboration, durability, centralized control and CI/CD use. Use appropriate locking/concurrency protection and encryption/access control.

Q: What is a module?

A: A reusable Terraform configuration that encapsulates a resource pattern with inputs and outputs.

Q: How do you manage environments?

A: Prefer reusable modules with environment-specific variable/configuration layers. Separate state boundaries where blast radius and lifecycle justify it.

Q: How do you review Terraform changes?

A: Run formatting/validation, security/policy checks, plan, peer review, then apply through controlled CI/CD. Protect production state and credentials.

Scenario: Terraform plan wants to replace a production resource

Scenario: A small configuration change produces a destructive replacement.

Interview approach: Stop the apply. Inspect the plan, identify which attribute forces replacement, check whether the real infrastructure drifted, review state and provider behavior, and determine whether the change is actually intended. Never approve a destructive production plan simply because Terraform generated it.

Useful checks:

```
terraform fmt -check
terraform validate
terraform plan
terraform state show <resource>
```

9. CI/CD + AWS

Q: Describe a production CI/CD flow for EKS.

A: Developer pushes code → CI builds/tests/scans → image is pushed to ECR with an immutable tag → image is signed → deployment manifests/Helm values are updated in Git → ArgoCD reconciles the cluster → health checks/observability validate rollout → rollback is available through Git.

Q: Why immutable image tags?

A: They provide traceability and avoid ambiguity caused by moving tags such as latest. A Git SHA or content digest maps a deployment to a specific artifact.

Q: Where should secrets live in CI/CD?

A: Use a managed secret system or workload identity. Avoid plaintext credentials in Git and avoid echoing secrets in pipeline logs.

Q: Jenkins controller vs agent?

A: The controller orchestrates jobs and manages Jenkins configuration; agents execute builds. Keep build workloads off the controller and use ephemeral/isolated agents where practical.

Q: Why GitOps?

A: The desired state is version-controlled in Git and a reconciler such as ArgoCD continuously compares actual state with desired state. It improves auditability, repeatability and rollback.

10. High Availability, DR & Architecture

Q: How would you design a highly available application on AWS?

A: Use multiple AZs, stateless application instances/pods, load balancing, autoscaling, Multi-AZ data services, resilient networking, backups, monitoring and automated deployment. Remove single-AZ and single-instance dependencies.

Q: RTO vs RPO?

A: RTO is how quickly service must be restored. RPO is how much data loss is acceptable measured in time.

Q: How would you design multi-AZ EKS?

A: Use worker capacity across at least two or three AZs, spread replicas using topology constraints/anti-affinity, use an appropriate load balancer, AZ-aware storage, resilient NAT/networking and PodDisruptionBudgets.

Q: What is a single point of failure?

A: A component whose failure causes service outage because there is no redundant path or replacement. Examples include a single AZ dependency, single NAT in a critical design, or a single non-redundant instance.

Q: How do you validate DR?

A: Backups alone are not DR. Test restoration, measure RTO/RPO, verify application dependencies, document runbooks and periodically execute recovery exercises.

11. AWS Cost Optimization

Q: How do you reduce EKS cost?

A: Right-size requests/limits and nodes, use autoscaling, use Spot for interruption-tolerant workloads, consolidate underutilized nodes, select appropriate instance families, control log retention and consider VPC endpoint/NAT patterns.

Q: How do you reduce NAT Gateway cost?

A: Use VPC endpoints for supported AWS services, reduce unnecessary cross-AZ traffic, and architect private workloads to avoid avoidable internet egress.

Q: How do you reduce S3 cost?

A: Lifecycle transitions, expiration policies, intelligent tiering where appropriate, compression and retention controls.

Q: What is right-sizing?

A: Matching compute/storage capacity to actual workload demand rather than provisioning excessive resources.

12. Security & DevSecOps Scenarios

Scenario: Image has a critical CVE

Scenario: A production image scan reports a critical vulnerability.

Interview approach: Confirm whether the vulnerable package is reachable/exploitable, identify a patched base image/package, rebuild and retest, block promotion based on policy where appropriate, sign the replacement image, deploy progressively and document the exception if an immediate patch is impossible.

Scenario: Pod is running as root

Scenario: A security review flags privileged container execution.

Interview approach: Use a non-root image/user, Kubernetes securityContext, drop unnecessary Linux capabilities, set read-only root filesystem where practical, restrict privilege escalation and enforce the baseline through admission policy.

Scenario: Developer requests AdministratorAccess for troubleshooting

Scenario: The request is urgent.

Interview approach: Do not grant permanent admin by default. Identify the exact operation, provide temporary least-privilege access or a controlled break-glass process, audit the action and revoke access after the incident.

13. Master Troubleshooting Playbooks

Symptom	First checks	Likely areas
ALB 5xx	Target health, Service endpoints, readiness	Ingress, SG, app port, pods
Pod Pending	describe pod, events	CPU/memory, taints, affinity, capacity
CrashLoopBackOff	logs --previous, describe	startup, config, secret, probe, dependency
ImagePullBackOff	describe pod	ECR, IAM, tag, DNS/network
Node NotReady	describe node, kubelet/containerd	runtime, disk, memory, CNI, network
RDS timeout	DNS + nc + SG	route, SG, NACL, DB state
EC2 unreachable	status + SG + route	IGW, public IP, NACL, host firewall
No internet from private subnets	route + NAT	NAT, route table, NACL, DNS
High CPU	top, HPA, requests	app load, limits, node sizing
Terraform destroy/replace	plan details	forced replacement, drift, state

Core AWS troubleshooting command set

```
# AWS
aws sts get-caller-identity
aws ec2 describe-instances
aws ec2 describe-security-groups
aws ec2 describe-route-tables
aws elbv2 describe-load-balancers
aws elbv2 describe-target-health
aws eks describe-cluster --name <cluster>
aws rds describe-db-instances

# Kubernetes
kubectl get nodes
kubectl get pods -A
kubectl get svc,ingress -A
kubectl get endpoints -A
kubectl describe pod <pod> -n <ns>
kubectl logs <pod> -n <ns> --previous
kubectl get events -A --sort-by=.lastTimestamp
kubectl top nodes
kubectl top pods -A
```

14. Architect-Level Scenarios You Should Be Ready For

Design a 50+ microservice platform on EKS.

Use multi-AZ VPC, private worker nodes, ALB ingress, ECR, managed node groups, autoscaling, workload identity, Helm/GitOps, centralized observability, policy enforcement, secure secrets, RDS where appropriate, resilient storage and CI/CD. Explain why each component exists rather than listing services.

Design an internet-facing API platform.

Route 53 → public ALB → private EKS workloads → private RDS. Use WAF where appropriate, TLS termination, restrictive SGs, autoscaling, health checks, logging/metrics and multi-AZ deployment.

Design a private EKS cluster.

Keep worker nodes and, where requirements justify it, the API endpoint private. Provide access from approved networks, use VPC endpoints for required AWS services, ensure DNS and routing are correct, and avoid depending on unrestricted internet access.

Design a secure CI/CD pipeline.

Jenkins agents build and test, run SAST/SCA/container/IaC scans, build immutable images, sign them, push to ECR, update GitOps manifests, and let ArgoCD deploy. Secrets come from managed identity/secret systems rather than source code.

Design for failure of one AZ.

Ensure load balancers, compute, pods and data services have cross-AZ redundancy. Avoid single-AZ NAT/storage dependencies, use topology-aware scheduling and verify that capacity remains after an AZ failure.

15. 100 Rapid-Fire Questions

Q: 1. What is an AZ?

A: An isolated location within an AWS Region.

Q: 2. What is a Region?

A: A geographic AWS deployment containing multiple AZs.

Q: 3. What is an ENI?

A: A virtual network interface attached to resources.

Q: 4. What is an Elastic IP?

A: A static public IPv4 address allocated to an AWS account.

Q: 5. What is Route 53?

A: AWS DNS and traffic-routing service.

Q: 6. What is health checking?

A: A mechanism to determine whether a target can serve traffic.

Q: 7. What is WAF?

A: Web Application Firewall for filtering HTTP/S requests.

Q: 8. What is Shield?

A: AWS DDoS protection service.

Q: 9. What is CloudFront?

A: AWS content delivery network.

Q: 10. What is ECR?

A: Managed container image registry.

Q: 11. What is ECS?

A: AWS managed container orchestration service.

Q: 12. ECS vs EKS?

A: ECS is AWS-native orchestration; EKS provides managed Kubernetes.

Q: 13. What is Fargate?

A: Serverless compute for containers where AWS manages underlying servers.

Q: 14. What is Lambda?

A: Event-driven serverless compute.

Q: 15. What is DynamoDB?

A: Managed NoSQL key-value/document database.

Q: 16. What is ElastiCache?

A: Managed in-memory caching service.

Q: 17. What is SNS?

A: Pub/sub notification service.

Q: 18. What is SQS?

A: Managed message queue.

Q: 19. Why SQS?

A: Decoupling and buffering between producers and consumers.

Q: 20. What is EventBridge?

A: Managed event bus and event-routing service.

Q: 21. What is Systems Manager?

A: Operations service for managed instances, automation, parameters and related capabilities.

Q: 22. What is Parameter Store?

A: Hierarchical parameter/configuration store.

Q: 23. What is KMS?

A: Managed key-management service.

Q: 24. What is CloudTrail?

A: AWS API auditing.

Q: 25. What is Config?

A: Resource configuration/compliance tracking.

Q: 26. What is GuardDuty?

A: Managed threat detection.

Q: 27. What is Security Hub?

A: Centralized security findings and posture management.

Q: 28. What is Inspector?

A: Automated vulnerability assessment service.

Q: 29. What is Macie?

A: Sensitive-data discovery/security service for S3.

Q: 30. What is a launch template?

A: Reusable EC2 launch configuration.

Q: 31. What is an AMI?

A: Machine image for EC2.

Q: 32. What is an instance type?

A: A compute/memory/network/storage capacity profile.

Q: 33. What is EBS snapshot?

A: Point-in-time backup of EBS data.

Q: 34. What is EFS?

A: Managed shared filesystem.

Q: 35. What is RDS?

A: Managed relational database service.

Q: 36. What is Aurora?

A: AWS cloud-optimized relational database engine compatible with MySQL/PostgreSQL.

Q: 37. What is backup retention?

A: How long automated backups are retained.

Q: 38. What is RPO?

A: Maximum acceptable data loss measured in time.

Q: 39. What is RTO?

A: Maximum acceptable recovery time.

Q: 40. What is horizontal scaling?

A: Adding more instances/pods.

Q: 41. Vertical scaling?

A: Increasing resources of an instance/pod.

Q: 42. What is HA?

A: Designing to continue service through component failure.

Q: 43. What is DR?

A: Recovering service after major failure/disaster.

Q: 44. What is blue/green?

A: Two environments; switch traffic between versions.

Q: 45. Canary?

A: Gradually expose a new version to a subset of traffic.

Q: 46. Rolling deployment?

A: Replace instances/pods progressively.

Q: 47. What is immutable infrastructure?

A: Replace rather than modify running infrastructure.

Q: 48. Why tagging?

A: Ownership, cost allocation, automation and governance.

Q: 49. What is AWS Organizations?

A: Central management of multiple AWS accounts.

Q: 50. What is SCP?

A: Organization-level permission guardrail.

Q: 51. Why multiple AWS accounts?

A: Isolation, security boundaries, billing and governance.

Q: 52. What is cross-account role assumption?

A: Temporary access to another account through an IAM role.

Q: 53. What is STS?

A: Service providing temporary security credentials.

Q: 54. What is MFA?

A: Additional authentication factor.

Q: 55. What is a trust policy?

A: Defines who/what can assume an IAM role.

Q: 56. What is an identity policy?

A: Permissions granted to an identity.

Q: 57. What is a bucket policy?

A: Resource policy controlling S3 bucket access.

Q: 58. What is encryption at rest?

A: Data encrypted while stored.

Q: 59. Encryption in transit?

A: Data protected during network transmission, commonly TLS.

Q: 60. What is TLS termination?

A: Decrypting TLS at a load balancer/proxy.

Q: 61. What is a private hosted zone?

A: Route 53 DNS zone associated with VPCs.

Q: 62. What is split-horizon DNS?

A: Different DNS answers based on client/network context.

Q: 63. What is CIDR overlap?

A: Two networks use overlapping IP ranges, complicating routing.

Q: 64. Why avoid CIDR overlap?

A: It prevents clean routing/peering and complicates connectivity.

Q: 65. What is a default route?

A: Catch-all route such as 0.0.0.0/0.

Q: 66. What is a NAT instance?

A: EC2-based NAT; older/self-managed alternative to managed NAT Gateway.

Q: 67. NAT Gateway HA?

A: Deploy per AZ for resilient architecture.

Q: 68. What is PrivateLink?

A: Private connectivity to services using interface endpoints.

Q: 69. What is VPC peering limitation?

A: Non-transitive connectivity.

Q: 70. What is TGW?

A: Central network transit hub.

Q: 71. What is Direct Connect?

A: Dedicated network connection from on-premises to AWS.

Q: 72. VPN?

A: Encrypted tunnel over internet to AWS.

Q: 73. What is hybrid cloud?

A: Integrated on-premises and cloud environment.

Q: 74. What is kubelet?

A: Kubernetes node agent.

Q: 75. What is kube-proxy?

A: Component supporting Service networking on nodes.

Q: 76. What is CNI?

A: Container Network Interface; provides pod networking.

Q: 77. What is CoreDNS?

A: DNS service used by Kubernetes.

Q: 78. What is HPA?

A: Horizontal Pod Autoscaler.

Q: 79. What is VPA?

A: Vertical Pod Autoscaler.

Q: 80. What is Cluster Autoscaler?

A: Adjusts node capacity based on scheduling needs.

Q: 81. What is Karpenter?

A: Kubernetes node provisioning/scaling system for cloud environments.

Q: 82. What is a taint?

A: Node restriction requiring matching toleration.

Q: 83. What is a toleration?

A: Allows a pod to schedule onto a tainted node.

Q: 84. What is a PDB?

A: PodDisruptionBudget limiting voluntary disruptions.

Q: 85. What is readiness probe?

A: Determines whether a pod should receive traffic.

Q: 86. Liveness probe?

A: Determines whether a container should be restarted.

Q: 87. Startup probe?

A: Gives slow-starting containers time before liveness/readiness checks take effect.

Q: 88. What is ConfigMap?

A: Non-secret configuration data.

Q: 89. What is Kubernetes Secret?

A: Kubernetes object intended for sensitive configuration; protect it with RBAC/encryption and avoid treating base64 as encryption.

Q: 90. What is RBAC?

A: Role-based authorization.

Q: 91. What is namespace?

A: Logical isolation/grouping within a Kubernetes cluster.

Q: 92. What is Helm?

A: Kubernetes package manager/template system.

Q: 93. What is ArgoCD?

A: GitOps continuous delivery/reconciliation tool.

Q: 94. What is Prometheus?

A: Metrics collection/querying system.

Q: 95. What is Grafana?

A: Visualization/observability dashboarding platform.

Q: 96. What is Loki?

A: Log aggregation system commonly paired with Grafana.

Q: 97. What is SCA?

A: Software Composition Analysis for dependency vulnerabilities.

Q: 98. What is SAST?

A: Static Application Security Testing.

Q: 99. What is IaC scanning?

A: Scanning infrastructure definitions for security/compliance issues.

Q: 100. What is image signing?

A: Cryptographically signing container images to verify provenance/integrity.

Q: 101. What is policy as code?

A: Machine-enforced rules for infrastructure or workloads.

Q: 102. Why GitOps?

A: Versioned desired state plus automated reconciliation.

Q: 103. What is drift?

A: Actual infrastructure differs from declared configuration.

Q: 104. What is idempotency?

A: Repeated execution produces the same intended state.

Q: 105. What is observability?

A: Ability to understand system internal state from outputs such as metrics, logs and traces.

16. 30-Question Mock Interview

Q: 1. Walk me through your AWS platform architecture.

A: Use a diagram: Route 53 → ALB → EKS → RDS, plus VPC, IAM, ECR, CloudWatch/Prometheus and CI/CD.

Q: 2. Why private subnets?

A: Reduce direct exposure and force controlled ingress/egress.

Q: 3. How does a private EKS node pull an image from ECR?

A: IAM authorization plus network/DNS connectivity to ECR, often through NAT or appropriate VPC endpoints.

Q: 4. How does a pod access S3?

A: Pod identity/IRSA with least-privilege role.

Q: 5. Why not attach S3 admin to node role?

A: Violates workload isolation and least privilege.

Q: 6. ALB returns 503. What do you do?

A: Target health → endpoints → readiness → ports → SG → application.

Q: 7. RDS times out. What do you do?

A: DNS → port → SG → routes → NACL → RDS state.

Q: 8. Node is NotReady. What do you do?

A: Conditions → kubelet/runtime → resources → CNI/network → recent changes.

Q: 9. How do you upgrade EKS?

A: Compatibility review → control plane → add-ons → nodes → rolling validation.

Q: 10. How do you secure EKS?

A: IAM, RBAC, workload identity, SG/network policy, admission policy, image security, secrets and logging.

Q: 11. How do you handle secrets?

A: Managed secret store/workload identity; never source-control or log secrets.

Q: 12. How do you make deployments safe?

A: Immutable artifacts, automated tests/scans, GitOps, progressive rollout, health checks and rollback.

Q: 13. How do you reduce AWS cost?

A: Right-size, autoscale, Spot where suitable, optimize NAT/logs/storage.

Q: 14. How do you handle an AZ outage?

A: Multi-AZ compute/load balancing/data and avoid single-AZ dependencies.

Q: 15. How do you design DR?

A: Define RTO/RPO, backups/replication, automated recovery and test restoration.

Q: 16. How do you investigate a latency spike?

A: Define scope/time, check ALB/app/db metrics, traces/logs, recent changes, dependency latency.

Q: 17. What is the difference between SG and NACL?

A: Stateful resource-level vs stateless subnet-level.

Q: 18. What is IAM trust policy?

A: Who can assume the role.

Q: 19. What is an S3 bucket policy?

A: Resource-based access policy.

Q: 20. Why VPC endpoints?

A: Private AWS service access and reduced internet/NAT dependency.

Q: 21. How does Terraform protect infrastructure consistency?

A: State + declarative configuration + plan/apply + code review.

Q: 22. What if Terraform wants to delete a production resource?

A: Stop, inspect plan/state/drift/provider behavior, correct configuration, then apply deliberately.

Q: 23. How do you manage multiple environments?

A: Reusable modules with separate state and controlled environment configuration.

Q: 24. How do you monitor EKS?

A: Prometheus/Grafana plus AWS CloudWatch and load-balancer/service metrics.

Q: 25. How do you troubleshoot a failed Jenkins deployment?

A: Pipeline logs → artifact/image → registry → manifest → ArgoCD sync → Kubernetes rollout.

Q: 26. Why ArgoCD if Jenkins can deploy?

A: Separate CI from CD and continuously reconcile desired state from Git.

Q: 27. How do you handle a critical CVE?

A: Assess, patch/rebuild, rescan, sign, promote and deploy; document exceptions.

Q: 28. What would you automate first on a new AWS platform?

A: VPC, IAM baseline, EKS, logging/monitoring, CI/CD templates and guardrails through Terraform.

Q: 29. What AWS area are you strongest in?

A: Be honest: emphasize the AWS components you have actually operated, especially EKS/VPC/IAM/EC2/load balancing.

Q: 30. What AWS area are you less experienced in?

A: Name one area honestly and explain your conceptual understanding and how you would safely learn/operate it.

17. Interview Traps — Avoid These Answers

“Private subnet means no internet.”

Incorrect. A private subnet can have outbound internet through NAT Gateway.

“Security groups are stateless.”

Incorrect. Security Groups are stateful; NACLs are stateless.

“Multi-AZ and Read Replica are the same.”

Incorrect. Multi-AZ is primarily HA/failover; replicas are primarily read scaling.

“Pod Running means application is healthy.”

Incorrect. Readiness, probes, endpoints and application health matter.

“Base64 Kubernetes Secret is encrypted.”

Incorrect. Base64 is encoding. Protect Kubernetes Secrets with RBAC and encryption at rest.

“NAT Gateway accepts inbound internet traffic.”

Not unsolicited inbound traffic. It supports outbound initiated connections from private resources.

“EKS control plane runs on our EC2 nodes.”

AWS manages the EKS control plane.

“Attach AdministratorAccess to solve IAM issues.”

Not a production solution. Diagnose the exact denied action and apply least privilege.

“Terraform state is just a cache.”

Incorrect. It is a critical mapping between configuration and managed resources.

“If the pod is healthy, ALB must be healthy.”

Incorrect. There are multiple layers: ingress, target registration, health checks, Service endpoints and readiness.

18. Final 60-Minute Revision Sheet

Memorize these architecture chains:

Internet → Route 53 → ALB → Target Group → EKS Service/Ingress → Pod

Pod → Node ENI → Private Subnet → NAT Gateway → IGW → Internet

Private workload → VPC Endpoint → AWS Service (for supported services)

Pod → ServiceAccount → EKS Pod Identity/IRSA → IAM Role → AWS API

Application → RDS endpoint → route → SG → RDS

Developer → Git → Jenkins CI → scan/test → ECR → GitOps repo → ArgoCD → EKS

Final interview rules

- Do not answer with a service name only. Explain the traffic/data/identity flow.
- If asked “why?”, give the trade-off.
- If asked troubleshooting, start from the symptom and move layer by layer.
- If you do not have direct production exposure, say so and give the correct design approach.
- Use your real EKS/CI/CD experience as the anchor. Connect AWS services to that platform.
- Do not overclaim 3 years of hands-on experience with every AWS service.
- Keep answers 30–90 seconds unless the interviewer asks for depth.
- Draw architecture verbally: DNS → LB → compute → data → identity → observability.

High-probability topics for this interview

VPC • Route Tables • SG/NACL • IGW/NAT • VPC Endpoints • ALB/NLB • IAM • EKS • ECR • Pod Identity/IRSA • EC2/ASG • RDS • EBS/EFS • S3 • CloudWatch • Terraform • HA/DR • Cost Optimization • Security • Production Troubleshooting